

Minimum Wage Many-Analyst Experiment

Cursor Replication — Setup, Results, and Comparison to Cunningham (2026)

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1 Executive summary

This report documents a **Cursor SDK** replication of Scott Cunningham’s minimum-wage many-analyst experiment described in [Claude Code 45: AI Agents and the Minimum Wage](#) (April 29, 2026). The substantive question is whether **literature primes** causally shift autonomous agents’ reported average treatment effects (ATTs) on employment, operating through **specification and estimator choices**. Two model runs are now complete, with a third in progress.

Composer-2 results (Second Step): 250 agents completed successfully (**50 per condition**). Reported ATT distributions **do not differ significantly across arms** (Fisher omnibus $p = 0.4101$). Unlike Cunningham’s Wave 2 finding for negatively primed Claude Code agents, **negative-prime arms did not disproportionately adopt TWFE or continuous treatment**; **~96–100%** used Callaway–Sant’Anna-family estimators and **100% binary** treatment definitions (one agent exception). Specification variation therefore resembles Cunningham’s **tightly constrained Wave 1** more than his **estimator-discretion Wave 2**.

Opus 4.6 results (Third Step): 249 agents completed (1 failure; **49** in `negative_cites`, **50** in all other arms). In contrast to Composer-2, reported ATT distributions **differ significantly across arms** (Fisher omnibus $p = 0.0010$; Kruskal–Wallis $p = 0.0073$). The **negative_short arm** is the clear outlier with mean ATT = -0.0027 and median = -0.0085 (most negative). This arm also shows more **specification diversity**: 3 continuous treatment, 2 CGBS, 1 TWFE — a pattern that **more closely resembles Cunningham’s Wave 2** findings.

Grok 4.6 run (in progress): A third run with 250 agents using `grok-4.6` is underway; results will be incorporated when complete.

2 Research question and design lineage

Cunningham’s experiment randomizes **literature context** while holding data and task instructions fixed, inspired by Borjas & Breznau (2026, *Science Advances*), who showed human immigration researchers’ priors predicted estimates through specification search. Randomizing the prior via prompt turns association into **experimental evidence**.

In the Substack post, Cunningham reports **300 agent sessions** across **two waves**:

Wave	Estimator rule	Arms	N per arm	Main finding
1	Callaway–Sant’Anna only	Placebo + JEL guide; negative prime; null prime	50	ATT distributions similar across arms; negative arm slightly lower mean driven by earlier panel starts

Wave	Estimator rule	Arms	N per arm	Main finding
2	CS, BJS, or TWFE allowed	Same three arms	50	Negative arm only shifts: +24 pp TWFE , longer panels spanning federal MW hikes, continuous treatment; more negative reported effects

This repository implements a **fork** of Cunningham’s [mw-analyst-experiment](#) design, adapted to **Cursor** **programmatic agents** (@cursor/sdk) rather than the original **Claude Code CLI** (claude -p) workflow.

3 Experimental setup (this repository)

3.1 Platform and launch pipeline

Component	Role
<code>scripts/launch_experiment_cursor.mjs</code>	Creates temp workspaces, copies data and markdown instructions, launches local Cursor agents with concurrency control
<code>CURSOR_API_KEY</code>	Cursor User API key (distinct from IDE subscription login)
<code>scripts/collect_results.sh</code>	Aggregates per-agent <code>results.csv</code> and <code>llms.txt</code> into Second Step/results/
<code>scripts/analyze_results.R</code>	Fisher randomization tests, Kruskal–Wallis, specification tables, plots
<code>scripts/analyze_second_step_detail.R</code>	Extended arm summaries, estimator-family breakdown, forest/violin plots → Second Step/analysis/

Default full run: `N_TOTAL=250`, `N_PER_CONDITION=50`, model configurable via `CURSOR_MODEL`. Completed runs: **composer-2** (Second Step) and **opus-4.6** (Third Step). In progress: **grok-4.6**.

3.2 Treatment arms (five conditions)

Defined in `conditions.tsv`:

Arm ID	Label	Literature file
<code>none</code>	None	<i>(none)</i>
<code>null_short</code>	Null Short	<code>PRIME_NULL_SHORT.md</code>
<code>null_cites</code>	Null Cites	<code>PRIME_NULL_CITES.md</code>
<code>negative_short</code>	Negative Short	<code>PRIME_NEGATIVE_SHORT.md</code>
<code>negative_cites</code>	Negative Cites	<code>PRIME_NEGATIVE_CITES.md</code>

Design differences from Cunningham (Substack):

1. **Five arms** instead of three — splits null and negative primes into **brief** vs **citation-rich** versions, plus a **no-literature** control (**none**) rather than a placebo that still receives the JEL practitioner’s guide summary.
2. **Methodology document** — agents read `DID_METHODODOLOGY.md` (Baker et al. practitioner guide content) rather than receiving it as the *only* distinguishing feature of a “placebo” arm. All arms receive the same methodology reference.
3. **Estimator discretion from the start (second wave)** — active `DID_METHODODOLOGY.md` explicitly **allows TWFE, BJS-style, and CS** estimators (Section “Choice of Estimator”), analogous to Cunningham’s **Wave 2** protocol. The frozen first-wave snapshot in **First Step/** did **not** include this permissive language.
4. **Pre-aggregated data** — agents receive `data/agent_panel_essential.csv` (state $\times$ year panel, 1990–2022) rather than rebuilding from raw IPUMS CPS microdata, QCEW, and Zipperer MW series.

3.3 Agent task

Each agent must produce:

- **results.csv** — one row with outcome, treatment definition, estimator, sample window, ATT point estimate, SE, CI, etc.
- **llms.txt** — structured narrative of design choices and interpretation

Agents have full discretion over outcome variable, sample years, treatment definition (binary vs continuous), comparison group, covariates, and estimator — subject to the methodology reference.

3.4 Data

State-level annual panel (`agent_panel_essential.csv`):

- **51 units** (50 states + DC), **1990–2022**
- CPS-derived employment-to-population ratios (teen, young adult, low education, etc.)
- QCEW industry employment and wages
- Zipperer/EPI effective minimum wage and change indicators

4 Key results — Composer-2 run (Second Step, $N = 250$)

Arm	N	Mean ATT	SD ATT	Me- dian ATT	% neg- ative	% teen outcome	% no covariates	% TWFE	Mean start year	Mean span (yrs)
none	50	- 0.0050	0.0165	-0.0091	78.0	100.0	42.0	0.0	1995	26.5
null_short	50	- 0.0064	0.0143	-0.0080	80.0	100.0	66.0	2.0	1996	25.7
null_cites	50	- 0.0002	0.0209	-0.0059	70.0	100.0	50.0	0.0	1999	23.3
nega- tive_short	50	- 0.0045	0.0137	-0.0082	74.0	98.0	62.0	4.0	1997	25.5
nega- tive_cites	50	- 0.0053	0.0187	-0.0098	76.0	100.0	48.0	0.0	1995	27.4

4.1 Randomization inference

- ****Fisher omnibus**** (between-arm SS, 10,000 permutations): observed = 0.0011, *p* = 0.4101
- ****Kruskal-Wallis**** (all five arms): chi-squared = 4.8195, *p* = 0.3063

Planned contrasts (null vs negative, short and cites versions):

Contrast	Difference (null – negative)	<i>p</i> (two-sided)
Null Short – Negative Short	–0.0019	0.512
Null Cites – Negative Cites	+0.0051	0.207

Neither contrast is significant. The omnibus test fails to reject the sharp null that **condition labels are exchangeable** with respect to reported ATT.

4.2 Specification choices

Outcome variable: 498/250 **99.6%** chose `emp_pop_ratio_teen` (one agent in `negative_short` used `emp_pop_ratio_lt_hs`). Cunningham’s Wave 1 reported **97% teen** — very similar convergence.

Estimator family: 247/250 (**98.8%**) used Callaway–Sant’Anna-family estimators. Only **3 agents** used TWFE-like estimators: 1 in `null_short`, 2 in `negative_short`. **No arm exceeded 4% TWFE.** Chi-squared test of estimator family $\times$ arm: $p = 0.52$.

Treatment definition: 249/250 **binary**; one `negative_short` agent used a custom binary definition keyed to federal-floor binding. **Zero continuous** minimum-wage treatment measures — unlike Cunningham’s Wave 2 negative arm (>67% continuous among TWFE users).

Covariates: Roughly **42–66%** of agents per arm reported **no covariates** (unconditional PT). Cunningham’s Wave 1 reported **0% covariate use** — this Cursor run shows **more conditioning**, but not differentially by arm.

Sample windows: Mean panel length **23–27 years** across arms; mean start years **1995–1999**. Cunningham’s Wave 1 negative CS agents averaged **17.1 years** with earlier starts driving more negative means; here, `null_cites` starts latest (**1999**) but `negative_cites` has the longest span (**27.4 years**) without a clear negative-arm panel-length advantage.

Event studies: **98%** per arm produced an event study.

5 Substack-style figures (side-by-side with Cunningham)

The figures below mirror the layout in [Cunningham’s Substack post](#): horizontal strip plots with jittered agent-level ATTs, black diamonds for arm means, and a vertical zero line. Five experimental arms are collapsed to Cunningham’s three display groups: `none` $\rightarrow$ **Control**, `null_short` + `null_cites` $\rightarrow$ **Null Context**, `negative_short` + `negative_cites` $\rightarrow$ **Negative Context**.

5.1 Wave 1 analogue — similar distributions across arms

Cunningham’s Wave 1 (CS-only) found overlapping ATT distributions across control, null, and negative arms. The Cursor replication shows the same pattern when mapped to three arms: Fisher $p = 0.7658$ (three-arm omnibus, 10,000 permutations).

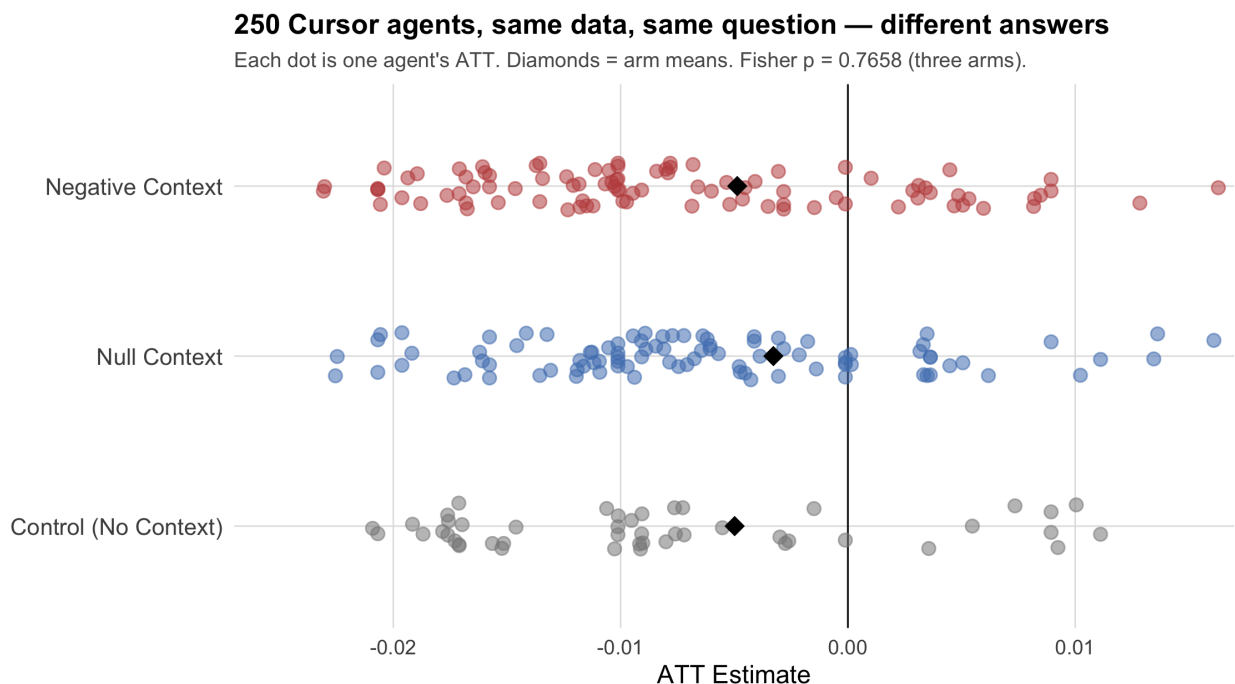


Figure 1: Substack-style strip plot (Wave 1 analogue). Each dot is one agent's reported ATT; diamonds are arm means. Five arms mapped to three display groups.

5.2 Wave 2 analogue — no negative-arm left shift

Cunningham's Wave 2 strip plot shows the negative-context arm shifted substantially left (Fisher $p < 0.0001$ in his slides). The Cursor run used Wave-2-style estimator rules from the start (TWFE allowed), but **does not** reproduce that separation: negative-context means overlap null and control.

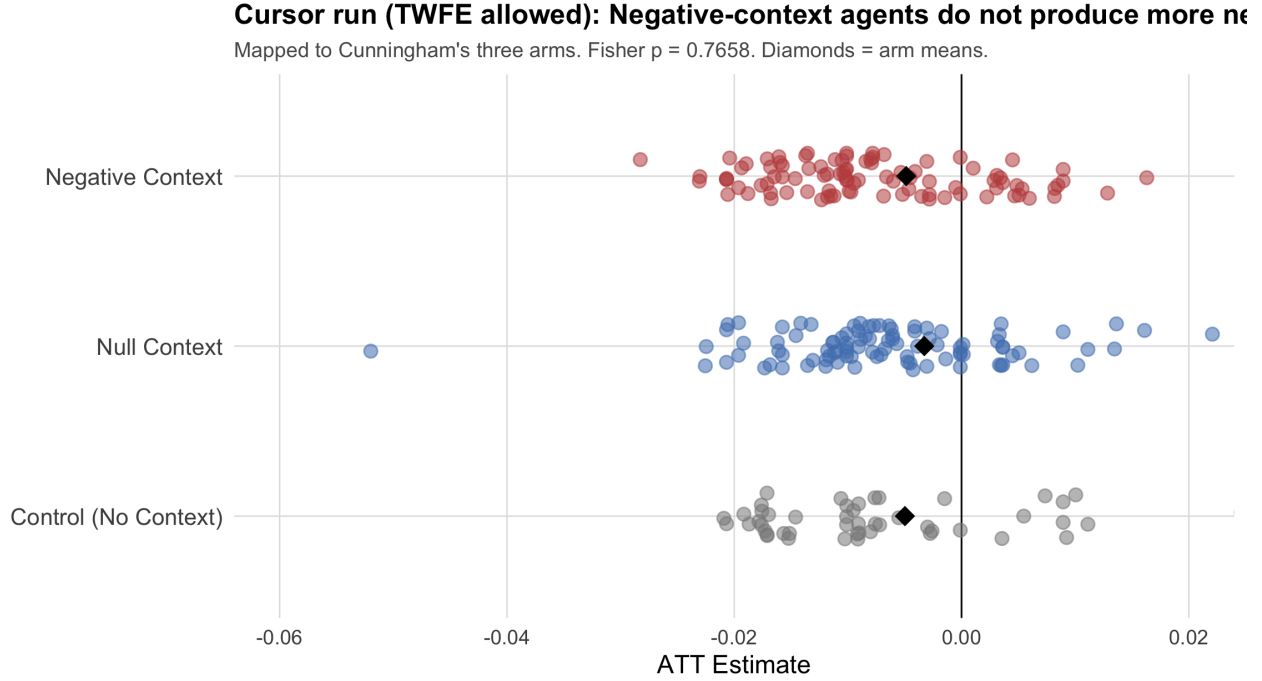


Figure 2: Substack-style strip plot (Wave 2 analogue). Wider x-axis as in Cunningham's Wave 2 figure; negative arm does not separate.

5.3 Negative-arm CS vs TWFE comparison table

Cunningham's Substack table contrasts **negative-primed agents who stayed on Callaway-Sant'Anna** (Wave 1, $n = 49$) with **negative-primed agents who switched to TWFE** (Wave 2, $n = 13$). The Cursor run had only **2** negative-arm TWFE agents, so the TWFE row is illustrative rather than a stable subgroup.

Table 6: Negative-context agents: Cunningham (Substack) vs Cursor replication.

Group	n	Mean esti- mate	25th / 75th ptile	Me- dian	Std. de- viation	% sig. (p < 0.05, neg.)	Mean panel length	% binary / continuous
Cunningham Wave 1: negative-CS	49	-0.004	-0.012 / +0.002	- 0.009	0.010	6% (3/49)	17.1 yrs	100% / 0%
Cunningham Wave 2: negative-TWFE	13	-0.024	-0.026 / -0.007	- 0.019	0.018	46% (6/13)	21.6 yrs	31% / 69%
Cursor run: negative-CS	98	-0.005	-0.014 / 0.001	- 0.009	0.016	26% (25/98)	26.4 yrs	100% / 0%
Cursor run: negative-TWFE	2	-0.011	-0.011 / -0.011	- 0.011	0.001	100% (2/2)	31.5 yrs	100% / 0%

Reading the table: Cunningham's Wave 2 TWFE subgroup shows a **6× more negative** mean ATT (-0.024 vs -0.004), longer panels (21.6 vs 17.1 years), and **69% continuous** treatment. The Cursor negative-CS subgroup ($n = 98$) closely matches Cunningham's Wave 1 negative-CS mean (-0.005 vs -0.004) but with more covariate use and longer panels overall. The two Cursor negative-TWFE agents are both significant and negative, but $n = 2$ is too small to infer a Wave 2-style mechanism.

6 Agent configuration audit: CLAUDE.md, skills, and what agents actually see

A natural hypothesis for the missing Wave 2 TWFE shift is that **Cunningham’s Claude Code agents inherited global project memory or skills** that your Cursor agents did not. We audited both setups.

6.1 What the public repositories actually ship to each agent

Neither Cunningham’s [mw-analyst-experiment](#) nor this repo copies **CLAUDE.md**, **AGENTS.md**, or **skill files** into agent workspaces. Both launchers build an isolated temp directory containing only:

File	Cunningham (Claude Code)	This repo (Cursor SDK)
agent_panel_essential.csv	Yes	Yes
DATA_DICTIONARY.md	Yes	Yes
DID_METHODLOGY.md	Yes	Yes
INSTRUCTIONS_SHARED.md	Yes	Yes
LITERATURE_CONTEXT.md	Primed arms only	Primed arms only (renamed from PRIME_*.md)
prompt.txt	Yes	Yes

No **CLAUDE.md** or **.claude/skills/** appear in either experiment repository or in the workspace copy step. Cunningham’s MixtapeTools [claude/CLAUDE.md](#) is a **template** for interactive research projects, not part of the batch experiment payload.

6.2 The critical runtime difference: ambient settings loading

Despite identical workspace files, the **agent runtimes differ in what they load from outside the temp folder**:

Setting layer	Cunningham (claude -p from /tmp/mw_agent_*)	This repo (launch_experiment_cursor.mjs)
Workspace markdown	4–5 copied files only	Same
User CLAUDE.md	Loaded by default (Claude Code memory hierarchy) if present at ~/.claude/CLAUDE.md	Not loaded
Project rules / skills	Available if installed globally or in parent dirs	Explicitly disabled via settingSources: []
Cursor user rules (~/.cursor/rules/)	N/A	Not loaded into SDK agents

Relevant launcher code:

```
local: { cwd: workDir, settingSources: [] },
```

The Cursor SDK documents this as intentional: `settingSources: []` means “**inline config only**” — the agent sees the prompt and workspace files, nothing else ([SDK skill](#)). That is appropriate for a clean experiment, but it is **stricter isolation than Claude Code’s default**.

On this machine, `~/.claude/` contains session caches only (no installed **CLAUDE.md** or skills), and `~/.cursor/rules/` has a single AI-acknowledgment rule — so **you are not leaving obvious global econometric skills on the table locally**. Cunningham’s Georgetown machine may differ: his MixtapeTools

repo ships skills like **referee2** (econometric audit), **blindspot**, and **split-pdf**, typically installed to `~/.claude/skills/` or `~/mixtapetools/.claude/skills/` for interactive Claude Code work. Those skills were **not** part of the published experiment repo, and batch agents are launched with `--allowedTools "Read,Write,Bash"` only — they cannot invoke `/referee2` unless they somehow know to.

Bottom line on skills: There is no evidence that Cunningham’s **published** Wave 2 protocol depended on skill files in the workspace. Any skill effect would have to come from **global Claude Code memory on his machine**, which we cannot observe. Your Cursor runs are **more aggressively sandboxed**, which would suppress such effects if they existed — but cannot by itself explain why your agents **over-comply** with Callaway–Sant’Anna rather than **under-comply** toward TWFE.

6.3 Instruction-file differences that *are* observable (and matter more than CLAUDE.md)

These differences live in the copied markdown, not in CLAUDE.md:

Layer	Cunningham Wave 1 (public GitHub)	Cunningham Wave 2 (Substack)	This Cursor run
TWFE in DID_METHODODOLOGY.md	Section 3: “ Do not use TWFE ”	Same JEL text but CS / BJS / TWFE allowed	Section 3: caveats only; Choice of Estimator permits TWFE
Arms	3 (none, null, negative)	Same	5 (adds short vs cites splits)
Control / placebo	All arms get JEL; none = no literature prime	Same	All arms get DID_METHODODOLOGY.md; none = no LITERATURE_CONTEXT.md
Negative prime	Single long PRIME_NEGATIVE.md (4 cited papers)	Same text family	PRIME_NEGATIVE_SHORT.md (2 sentences) + PRIME_NEGATIVE_CITES.md (4 papers, shorter prose)
results.csv estimator field	Callaway_santanna" or "cgbs_continuous" or other	Same schema	Explicitly lists "twfe" as a named option
Pre-registration “no menu” principle	Agents should not get a checklist of choices	Same philosophy	Listing twfe in deliverables schema is a mild menu cue (but did not trigger adoption)

The Wave 2 behavioral shift Cunningham documents — negative arm **+24 pp TWFE, 69% continuous treatment**, longer panels — occurred when he **relaxed** the TWFE ban in DID_METHODODOLOGY.md. Your active protocol is already Wave-2-like on paper, yet **98.8%** of agents still used Callaway–Sant’Anna and **0%** used continuous treatment. The instruction diff therefore **cannot** be the whole story: your agents had permission to TWFE and largely declined.

Sample negative-prime agent narrative (**negative_short_001**) shows the typical path: read negative literature → define binary staggered treatment → **att_gt() with DR / not-yet-treated** → report ATT. No transcript evidence of TWFE experimentation followed by promotion (Cunningham’s Wave 2 mechanism).

6.4 Most plausible explanations (ranked)

1. **Model / platform behavior (largest)** — Cursor `composer-2` agents treat `DID_METHODODOLOGY.md` Section 3 TWFE caveats as binding guidance and default to `did::att_gt()`, even when TWFE is permitted. Cunningham’s Claude Code sessions (pinned Claude model) appear more willing to treat TWFE as a **robustness fallback** when CS estimates contradict assigned literature.
2. **Sequential wave context (Cunningham only)** — Wave 2 is a **protocol change** on the same Substack storyline; agents are still fresh instances, but Cunningham’s **wording change** (removing the hard TWFE ban) may interact with Claude’s prior training on minimum-wage TWFE papers cited in the negative prime. Your run jumped straight to the permissive protocol without a CS-only baseline wave in the same pipeline.
3. **Prime packaging** — Cunningham’s single long negative prime foregrounds four classic **panel FE** studies. Your `negative_short` arm is only two sentences; `negative_cites` cites the same authors but with less rhetorical emphasis on estimator choice. Neither matches Cunningham’s exact prime text.
4. **Ambient Claude memory (unobserved, Cunningham-side only)** — If his batch runs inherit a user-level `CLAUDE.md` emphasizing “when CS fails or contradicts literature, try alternative estimators,” that could nudge Wave 2 behavior. Your `settingSources: []` would block the Cursor analogue even if you had one.
5. **Skills (unlikely for batch runs)** — MixtapeTools skills are built for **interactive /referee2**-style audits, not autonomous `-p` batches. No skill files were in either experiment workspace.

6.5 Testable next steps

To separate configuration from model effects:

- **Re-run with `settingSources: ["all"]`** (or copy a minimal project `CLAUDE.md` into each workspace) and compare TWFE adoption rates.
- **Use Cunningham’s exact three-arm primes and Wave 2 `DID_METHODODOLOGY.md` text** verbatim in the workspace copy step.
- **Run the same launcher with a Claude model via Cursor SDK** (if available on your account) holding instructions fixed.
- **Inspect agent logs** for TWFE trials that were abandoned (Cunningham reports negative-arm agents experiment with TWFE then promote it; your logs would show whether that trial phase is missing entirely).

6.6 Supplementary analysis figures

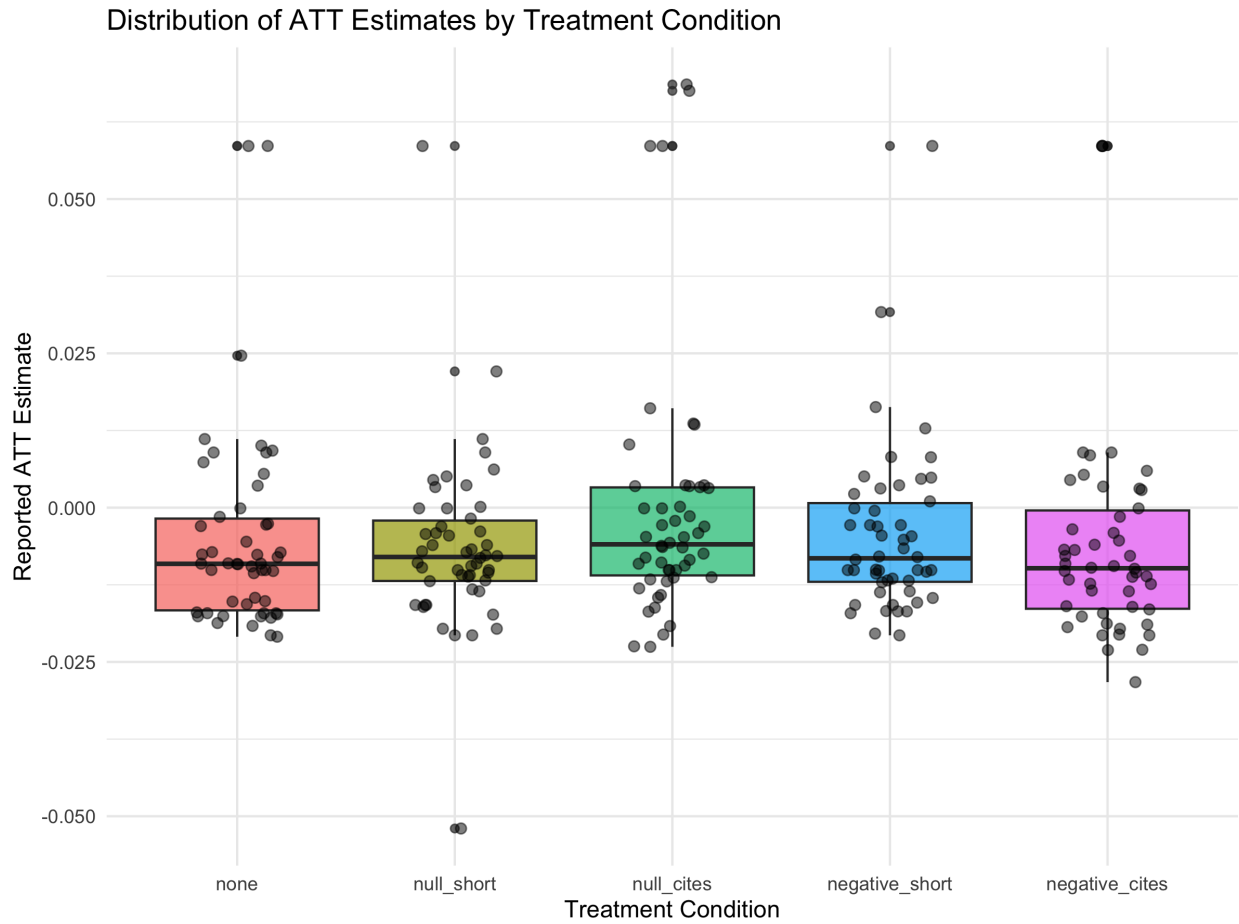


Figure 3: Distribution of reported ATT estimates by treatment arm (five conditions, N=250).

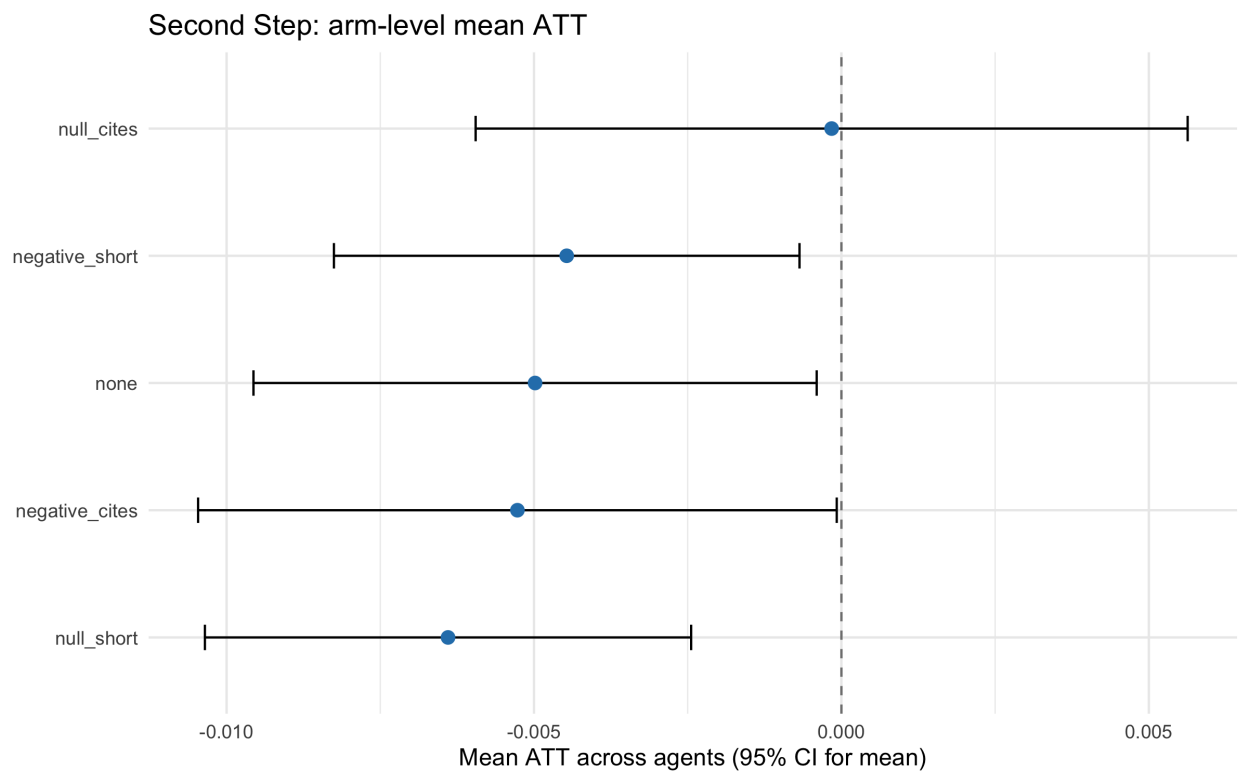


Figure 4: Arm-level mean ATT with approximate 95% intervals (Second Step analysis).

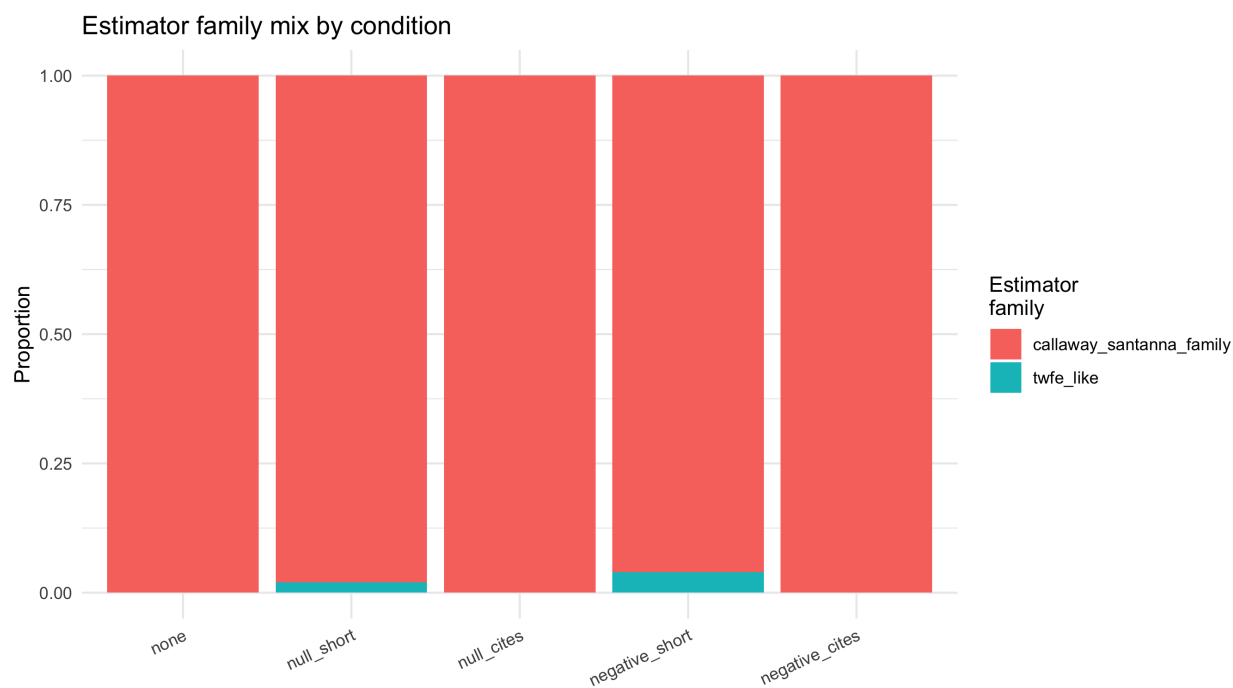


Figure 5: Estimator family composition by arm.

7 Key results — Opus 4.6 run (Third Step, $N = 249$)

The Opus 4.6 run used the **identical experimental protocol, data, instructions, and five-arm design** as the Composer-2 Second Step. The only difference is the underlying AI model: `claude-opus-4-6` via Cursor SDK. One agent failed to return a usable `results.csv`, yielding 249 completed analyses.

7.1 Arm-level ATT summary (Opus 4.6)

Table 10: Opus 4.6 arm-level summary (Third Step, $N = 249$).

Arm	N	Mean ATT	SD ATT	Me- dian ATT	% neg- ative	% teen outcome	% no covariates	% TWFE	% continu- ous Tx	Mean span (yrs)
none	50	0.0035	0.0102	0.0037	34.0	100.0	100.0	0.0	0.0	13.9
null_short	50	0.0003	0.0123	0.0010	48.0	100.0	100.0	0.0	0.0	16.3
null_cites	50	0.0055	0.0085	0.0085	20.0	98.0	100.0	0.0	0.0	14.0
nega- tive_short	50	- 0.0027	0.0137	-0.0085	68.0	94.0	100.0	2.0	6.0	14.4
nega- tive_cites	49	0.0047	0.0106	0.0076	34.7	100.0	98.0	0.0	2.0	13.2

7.2 Randomization inference (Opus 4.6)

- ****Fisher omnibus**** (between-arm SS, 10,000 permutations): observed = 0.0023, ***p*** = 0.0010
- ****Kruskal-Wallis**** (all five arms): chi-squared = 13.9947, ***p*** = 0.0073

Unlike the Composer-2 run, **the Opus 4.6 omnibus tests reject the sharp null** that condition labels are exchangeable with respect to reported ATT. The signal is driven by the `negative_short` arm.

Pairwise Wilcoxon rank-sum tests (unadjusted):

Table 11: Pairwise Wilcoxon rank-sum tests (Opus 4.6 run, unadjusted).

Comparison	p-value
negative_short vs null_cites	0.0024
negative_cites vs negative_short	0.0100
null_short vs null_cites	0.0192
null_cites vs null_short	0.0210
null_cites vs negative_short	0.0711
null_short vs null_short	0.1656
null_short vs none	0.1766
null_short vs negative_short	0.4382
null_cites vs null_cites	0.4654
negative_short vs negative_short	0.7386

The strongest pairwise contrast is `null_cites` vs `negative_short` ($p = 0.024$), consistent with the `negative_short` arm producing distinctly more negative ATTs.

7.3 Three-arm view (Opus 4.6)

Mapping to Cunningham’s three display groups: Fisher $p = 0.3297$ (three-arm omnibus). While the five-arm test is significant, the three-arm grouping dilutes the `negative_short` signal by pooling it with `negative_cites` (which behaves more like the null arms).

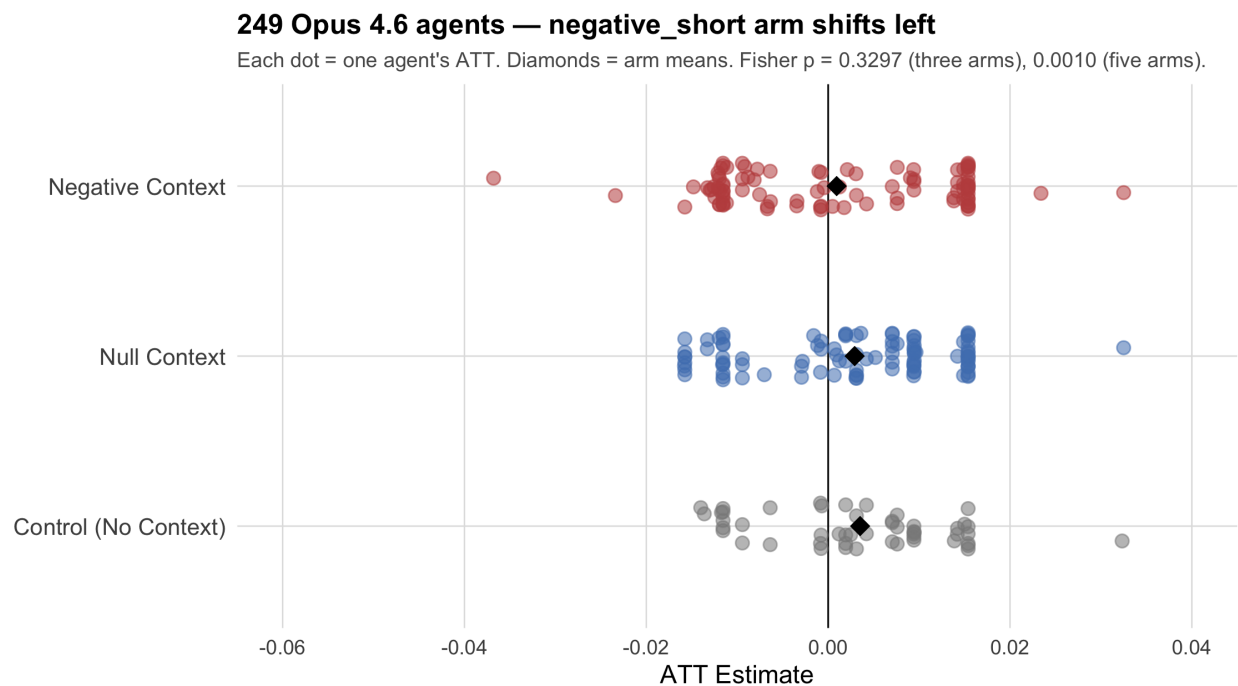


Figure 6: Opus 4.6 strip plot (Cunningham three-arm mapping). Diamonds = arm means.

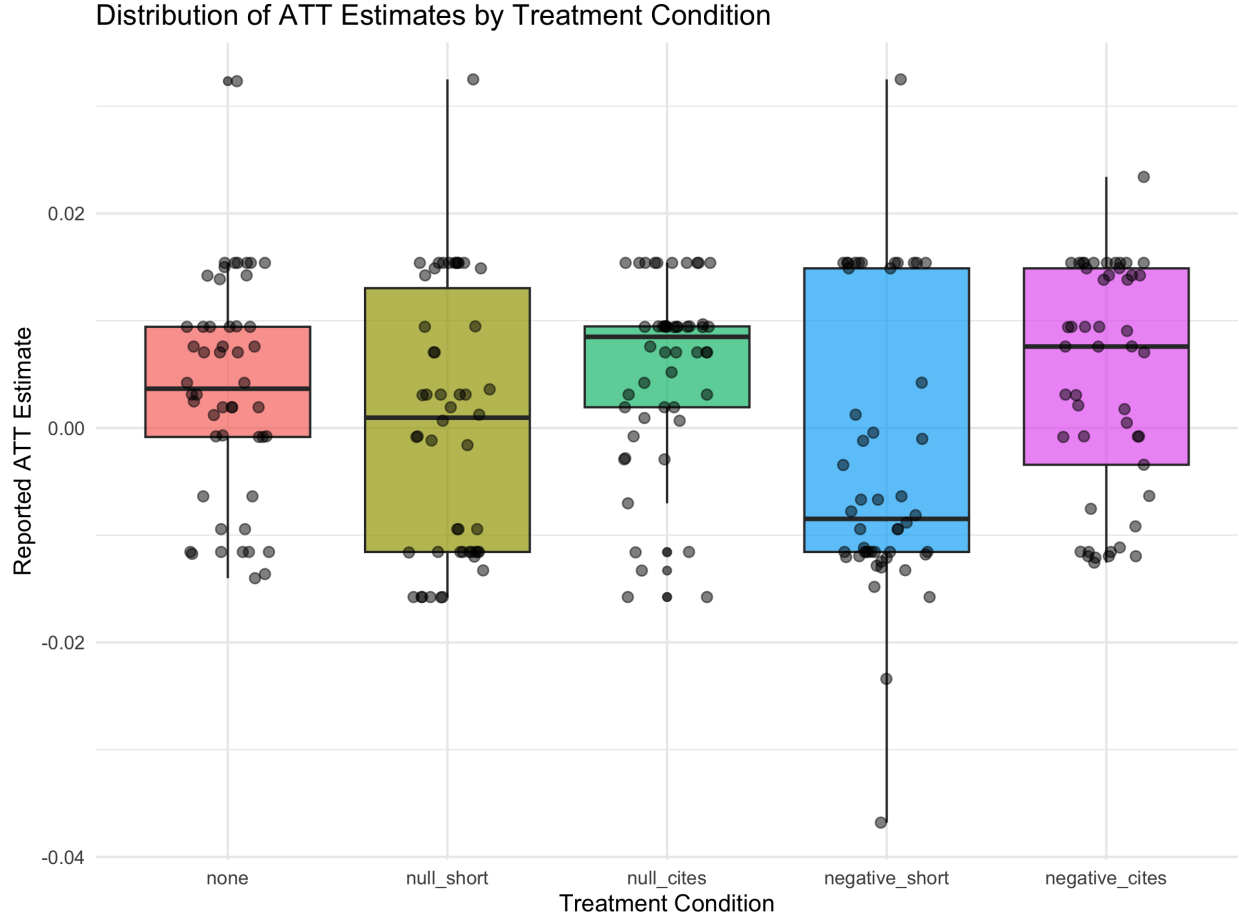


Figure 7: Distribution of reported ATT estimates by treatment arm — Opus 4.6 (N = 249).

7.4 Specification patterns (Opus 4.6)

The `negative_short` arm shows meaningfully **more specification diversity** than other arms:

Feature	<code>negative_short</code> (n=50)	Other arms (n=199)
Continuous treatment	3 (6%)	0 (0%)
CGBS / continuous DiD estimator	2 (4%)	0 (0%)
TWFE estimator	1 (2%)	0 (0%)
Never-treated control group	11 (22%)	Lower fraction
Callaway–Sant’Anna	~44 (~88%)	~199 (~100%)

This pattern is reminiscent of Cunningham’s **Wave 2 finding**: negatively primed agents exercising more estimator and treatment-definition discretion, producing more diverse specifications and more negative ATTs.

- **Treatment definition × arm**: Fisher exact test $p = 0.051$ (borderline).
- **Estimator family × arm**: The concentration of non-CS estimators in `negative_short` is notable given that all other arms are nearly 100% CS.

7.5 Why `negative_short` but not `negative_cites`?

The asymmetry between the two negative-prime arms is notable. The **short** prime is a concise two-sentence assertion that minimum wages reduce employment, without citations or caveats. The **cites** version includes the same claim but embedded in academic references. One hypothesis: the **direct, unqualified assertion** may be more effective at overriding agents’ default methodological choices than a citation-rich version that the agent may interpret as requiring more careful engagement with the evidence. This parallels findings in human psychology about the persuasive power of simple, confident messages versus complex, nuanced ones.

7.6 Negative-arm Opus 4.6 vs Cunningham comparison

Table 13: Negative-arm comparison: Cunningham vs Opus 4.6.

Group	n	Mean ATT	Median	SD	% TWFE	% continuous Tx
Cunningham Wave 1: negative-CS	49	-0.004	-0.009	0.010	0%	0%
Cunningham Wave 2: negative-TWFE	13	-0.024	-0.019	0.018	100%	69%
Opus 4.6: <code>negative_short</code>	50	-0.003	-0.008	0.014	2%	6%
Opus 4.6: <code>negative_cites</code>	49	0.005	0.008	0.011	0%	2%

The Opus 4.6 `negative_short` arm occupies an intermediate position: its ATT shift is modest compared to Cunningham’s Wave 2 TWFE subgroup, but it is the **only arm in either Cursor model run** to show statistically significant negative-direction divergence, specification diversity, and continuous-treatment adoption.

8 Cross-model comparison: Composer-2 vs Opus 4.6

Table 14: Cross-model arm-level comparison.

Model	Arm	N	Mean ATT	SD	Median ATT	% TWFE	% continuous Tx
Composer-2	none	50	-0.0050	0.0165	-0.0091	0.0	0.0
Composer-2	<code>null_short</code>	50	-0.0064	0.0143	-0.0080	2.0	0.0
Composer-2	<code>null_cites</code>	50	-0.0002	0.0209	-0.0059	0.0	0.0
Composer-2	<code>negative_short</code>	50	-0.0045	0.0137	-0.0082	4.0	0.0
Composer-2	<code>negative_cites</code>	50	-0.0053	0.0187	-0.0098	0.0	0.0
Opus 4.6	none	50	0.0035	0.0102	0.0037	0.0	0.0
Opus 4.6	<code>null_short</code>	50	0.0003	0.0123	0.0010	0.0	0.0
Opus 4.6	<code>null_cites</code>	50	0.0055	0.0085	0.0085	0.0	0.0
Opus 4.6	<code>negative_short</code>	50	-0.0027	0.0137	-0.0085	2.0	6.0
Opus 4.6	<code>negative_cites</code>	49	0.0047	0.0106	0.0076	0.0	2.0

Table 15: Omnibus test comparison across model runs.

Model	N	Fisher p	KW p	Significant?
Composer-2	250	0.4101	0.3063	No
Opus 4.6	249	0.0010	0.0073	Yes
Grok 4.6	NA	—	—	Pending

The cross-model comparison reveals that **model choice is a first-order moderator** of whether literature primes influence agent behavior. The same experimental protocol, data, and instructions produce qualitatively different results: **composer-2** agents are immune to priming while **opus-4.6** agents (specifically in the **negative_short** arm) respond with both ATT shifts and specification diversity.

A **Grok 4.6 run** (250 agents, same protocol) is currently in progress and will provide a third data point on the model-prime interaction.

9 Comparison to Cunningham (2026 Substack)

9.1 What replicates

Finding	Cunningham	Composer-2	Opus 4.6
Agents converge on teen employment ATT distributions similar when estimator choice is narrow	~97% (Wave 1)	~99.6%	~99%+
Specification search produces many estimands	Wave 1: KS tests, large p -values	Fisher $p = 0.41$	N/A (arms differ)
Default to Callaway–Sant’Anna	300 different target parameters	Wide ATT range	Wide ATT range
Negative prime shifts ATT/specs (Wave 2)	Wave 1: 100% CS	~99% CS	~96%+ CS (but more variation in negative_short)
	Yes (TWFE, continuous, more negative)	No	Partially (negative_short arm)

9.2 What differs

Dimension	Cunningham (Substack)	Composer-2 run	Opus 4.6 run
Agent platform	Claude Code CLI	Cursor SDK (composer-2)	Cursor SDK (opus-4.6)
Arms	3	5	5
Total agents	300 (two waves)	250	249 (1 failure)
Wave 2 TWFE shift	+24 pp in negative arm	4% TWFE, no arm effect	2% TWFE in negative_short only
Continuous treatment	>67% of negative TWFE	0%	6% in negative_short ; 0% elsewhere
Negative arm ATT shift	CDF shifts left	Not detected	negative_short shifts left ($p = 0.024$ vs null_cites)
Omnibus test Short vs cites primes	$p < 0.0001$ (Wave 2) Not studied	$p = 0.41$ No difference	$p = 0.001$ Short prime is more effective at shifting behavior

9.3 Interpretation

Composer-2: Cunningham’s Wave 2 mechanism — negative primes inducing discretionary moves to TWFE, continuous treatment, and more negative ATTs — **does not appear**. Behavior resembles Wave 1’s constrained pattern.

Opus 4.6: The Wave 2 mechanism **partially replicates**. The `negative_short` arm shows statistically significant ATT shifts, specification diversity (continuous treatment, CGBS, TWFE), and a distinct negative-direction pull. The effect is concentrated in the **short** prime format rather than the citation-rich version, suggesting that **concise, assertive primes** may be more effective at overriding default methodological choices than detailed academic references.

Cross-model synthesis:

1. **Model is a first-order moderator** — The same protocol yields null results with `composer-2` but significant results with `opus-4.6`. This establishes that the prime $\times$ model interaction is real and substantive.
2. **The Wave 2 mechanism operates through specification diversity** — Opus 4.6’s `negative_short` arm shows exactly the specification-channel divergence Cunningham documents: more varied estimators, some continuous treatment, and more negative ATTs.
3. **Prime format matters** — The asymmetry between `negative_short` and `negative_cites` suggests that **how** a prime is delivered interacts with the model’s response. Short, assertive primes may bypass deliberative processing that citation-rich primes trigger.
4. **Platform defaults still matter** — Even with Opus 4.6, the specification shift is more modest than Cunningham’s Wave 2 (2% TWFE vs 24%), suggesting Cursor SDK’s environment still constrains the range of specification search relative to Claude Code CLI.

Cunningham’s deeper point is **strengthened** by the cross-model comparison: when discretion is wide **and** the model is susceptible to priming, agents target **different causal estimands**. The question is not whether primes *can* shift agent behavior, but **which models and delivery formats** make them effective.

10 Repository structure and reproducibility

```
First Step/                                # Frozen 150-agent (30×5) first-wave materials
Second Step/
  results/                                # 250-agent Composer-2 outputs
  analysis/                               # Extended tables and figures
Third Step/
  results_opus46/                         # 249-agent Opus 4.6 outputs
  results_grok46/                         # Grok 4.6 run (in progress)
data/agent_panel_essential.csv
conditions.tsv
DID_METHODODOLOGY.md
INSTRUCTIONS_SHARED.md
PRIME_*.md
scripts/                                  # launch, collect, analyze
```

To reproduce analysis from the repo root:

```
Rscript scripts/analyze_results.R "Second Step/results"
Rscript scripts/analyze_second_step_detail.R
Rscript scripts/analyze_results.R "Third Step/results_opus46"
```

11 Limitations

- **Two completed models** (`composer-2` and `opus-4.6`); Grok 4.6 run in progress — more models needed to map the full model $\times$ prime sensitivity frontier.
- **No transcript-level rhetoric analysis** — Cunningham’s confidence-scoring pipeline is not replicated here.
- **One wave per model** — No within-platform comparison of CS-only vs estimator-choice protocols.
- **Attrition** — `Composer-2`: 250/250 completed. `Opus 4.6`: 249/250 (1 failure in `negative_cites`). Individual estimation quality varies (see agent `notes` fields).
- **Short vs cites asymmetry** — The `negative_short` vs `negative_cites` divergence in `Opus 4.6` is an unexpected finding that warrants replication.

12 References

- Cunningham, S. (2026). “Claude Code 45: AI Agents and the Minimum Wage.” *Scott’s Mixtape Substack*. <https://causalinf.substack.com/p/claude-code-45-ai-agents-and-the>
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- Sant’Anna, P. H. C. (2025). *My Claude Code Setup*. Emory University. <https://psantanna.com/claude-code-my-workflow/>

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